

07 - Prior Preterm Birth and 17-OHPC

Companion reading handout for simultaneous listening.

Course: Medical Prevention of Spontaneous Early Delivery

Prior spontaneous preterm birth is one of the strongest clinical risk factors in obstetrics, but it is not the same phenotype as asymptomatic short cervix.

That distinction became more important after the rise and fall of seventeen alpha hydroxyprogesterone caproate, often spoken as 17-OHPC.

Meis

The original anchor was the Meis trial, published in the New England Journal of Medicine in two thousand three. The population was women with singleton pregnancy and a prior spontaneous preterm delivery. The regimen was two hundred fifty milligrams of intramuscular 17-OHPC weekly, beginning at sixteen to twenty weeks and continuing until delivery or thirty-six weeks.

The result looked clinically powerful. Recurrent preterm delivery before thirty-seven weeks occurred in thirty-six point three percent of the 17-OHPC group and fifty-four point nine percent of the placebo group. Before thirty-five weeks, the rates were twenty point six percent versus thirty point seven percent. Before thirty-two weeks, eleven point four percent versus nineteen point six percent.

Store the Meis memory correctly.

This was not vaginal progesterone for short cervix. It was weekly intramuscular 17-OHPC for the recurrence phenotype. The trial result explains why the field became enthusiastic. A high-risk population, a simple weekly regimen, and large apparent reductions in recurrent preterm birth.

PROLONG

But medicine does not end with the first positive trial. The question is whether the result replicates in the population for which the drug is used.

The PROLONG trial was the confirmatory trial. It randomized women with singleton pregnancy and prior spontaneous preterm birth to the same general regimen: two hundred fifty milligrams of 17-OHPC or placebo, weekly, continued until thirty-six weeks or delivery.

The result was not a softer version of Meis. It was different. Preterm birth before thirty-five weeks was eleven point zero percent with 17-OHPC and eleven point five percent with placebo, relative risk zero point nine five, confidence interval zero point seven one to one point two six. The neonatal morbidity index was five point six percent versus five point zero percent, relative risk one point one two, confidence interval zero point seven zero to one point six six.

Store that contrast as a discipline in evidence memory. Meis was thirty-six versus fifty-five for birth before thirty-seven. PROLONG was eleven versus eleven point five for birth before thirty-five. The baseline event rate changed dramatically, and the treatment signal did not replicate.

This is not a story about embarrassment. It is a story about evidence discipline.

The original trial asked an important question and found an apparent benefit. The confirmatory trial asked whether the benefit held in a broader regulatory setting and did not confirm it. The Food and Drug Administration withdrew approval of Makena and its generics. ACOG then updated its guidance: intramuscular 17-OHPC is not recommended for the primary prevention of preterm birth in patients with a history of spontaneous preterm birth.

Do Not Substitute Automatically

Now comes the common error.

If 17-OHPC is no longer recommended, can we simply substitute vaginal progesterone for anyone with a prior spontaneous preterm birth?

No.

ACOG states that vaginal progesterone may be considered for patients with a history of preterm birth, singleton gestation, and a shortened cervix. But vaginal progesterone has not been proven effective in the absence of a shortened cervix and should not be considered an alternative to 17-OHPC.

That sentence must be stored exactly.

Vaginal progesterone can re-enter the discussion when the patient has a shortened cervix. But then the indication is not merely prior preterm birth. The indication has become prior preterm birth plus a current short-cervix signal. The evidence object has changed.

Recurrence Risk

This is where the recurrence-risk data matter.

McManemy showed that prior birth history is not just yes or no. In a third-birth cohort, the preterm delivery risk was about five percent after two prior term births, thirteen percent after a preterm first birth followed by a term second birth, twenty-one percent after a term first birth followed by a preterm second birth, and forty-two percent after two prior preterm births. When both prior preterm births were very preterm, between twenty-one and thirty-one weeks, the recurrence risk reached fifty-seven percent.

Frequency, order, severity. That is the memory.

A patient with two prior very early spontaneous preterm births is not the same probability object as a patient with one prior late-preterm birth followed by a term delivery. Both may carry the phrase "prior preterm birth," but the baseline risk is different.

When Short Cervix Changes the Phenotype

Now add cervical length.

If the patient with prior spontaneous preterm birth develops a short cervix, especially before twenty-four weeks, the current pregnancy has a second risk signal. At that point, management may include nonmedical options such as cerclage, which are outside this course, and medical options such as vaginal progesterone depending on the exact phenotype and guideline context. But the resident must not use vaginal progesterone as a vague replacement for withdrawn intramuscular therapy.

The EPPPIC individual-participant-data meta-analysis reinforces both the promise and the danger of broad categories. In high-risk singleton pregnancies, preterm birth before thirty-four weeks was reduced with vaginal progesterone, relative risk zero point seven eight, confidence interval zero point six eight to zero point nine zero. For 17-OHPC the point estimate was also favorable, zero point eight three, but the confidence interval crossed one, from zero point six eight to one point zero one. EPPPIC also emphasized that absolute risk reduction is larger when the underlying risk is larger, especially with short cervix.

But EPPPIC grouped several high-risk phenotypes, and later recurrent-preterm-birth analyses asked a sharper question. Conde-Agudelo and Romero examined vaginal progesterone specifically for singleton pregnancy with prior spontaneous preterm birth. The overall pooled result looked beneficial, but that was driven by small studies. In the large studies, vaginal progesterone did not reduce preterm birth before thirty-seven weeks: relative risk zero point nine eight. It did not reduce preterm birth before thirty-four weeks: relative risk zero point nine four. When analysis was restricted to low-risk-of-bias studies, the results remained non-significant.

That is the correct internal map. Progestogens are not one interchangeable class. The evidence differs by drug, route, population, study quality, and endpoint. Vaginal progesterone has stronger evidence in sonographic short cervix than in prior preterm birth without short cervix. 17-OHPC lost support after the confirmatory trial and regulatory review.

This module should leave you with a mature evidence posture.

Do not ridicule Meis. Understand why it changed practice. Do not ignore PROLONG. Understand why confirmatory evidence matters. Do not replace 17-OHPC with vaginal progesterone automatically. Understand that a short cervix changes the phenotype, and only then changes the medical conversation.

The final memory sentence is this:

Meis was weekly intramuscular 17-OHPC for prior spontaneous preterm birth, with under-thirty-seven recurrence reduced from fifty-four point nine to thirty-six point three percent. PROLONG was the confirmatory trial, and under-thirty-five birth was eleven point zero versus eleven point five percent. Current guidance does not recommend 17-OHPC, and vaginal progesterone is not a substitute unless a shortened cervix creates a different evidence phenotype.

That is the disciplined way to carry this history into practice.